

Evaluating Discriminant Analysis and KNN for Multi-class Astronomical Object Classification

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Abstract—We compare three supervised learning methods for astronomical object classification using Sloan Digital Sky Survey (SDSS) photometric data: Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), and K-Nearest Neighbors (KNN). The dataset contains 92303 objects (galaxies, stars, quasars) described by five optical magnitudes and redshift. After preprocessing, LDA, QDA, and KNN achieved accuracies of 83%, 95%, and 97%, respectively. Our findings demonstrate that nonlinear decision boundaries—whether from relaxed parametric assumptions (QDA) or distance-based methods (KNN)—substantially outperform linear approaches. QDA exploits statistical structure when assumptions hold, while KNN adapts to local data geometry with superior accuracy. These results confirm the continued effectiveness of classical statistical and non-parametric methods for multi-class astronomical classification problems.

Index Terms—Astronomical classification, Discriminant analysis, KNN.

I. INTRODUCTION

Galaxies, stars, and quasars are fundamental astronomical objects, representing distinct physical phenomena in the Universe. Galaxies are large systems composed of billions of stars, gas, and dust, all gravitationally bounded. Quasars, also known as quasi-stellar objects (QSOs), correspond to extremely luminous galactic nuclei whose energy is powered by the accretion of matter onto supermassive black holes [1]. In contrast, stars are luminous bodies primarily composed of hydrogen and helium, where nuclear fusion processes generate energy and synthesize the chemical elements that contribute to the enrichment of the Universe [2].

In this analysis, we use a dataset derived from the SDSS Data Release 17 (DR17), one of the most important astronomical surveys available. The dataset provides photometric measurements obtained through five optical filters (u, g, r, i, z), as well as redshift information, which together are used as our input features for classification. Based on these data, we investigate the performance of different classification models for distinguishing between galaxies, stars, and quasars. Classification, in this context, refers to a supervised learning task in which a model learns to assign each object, described by a set of photometric predictors, to a predefined label representing one of the astronomical object.

Redshift and photometric bands are the most important variables for classifying galaxies, stars, and quasars, though they

contribute unequally to class discrimination [3]. Chatterjee and Ghosh [3] demonstrated through univariate analyses that redshift exhibits distributions with clearly distinct mean values and standard deviations across object classes, establishing it as the single most critical feature for classification, despite some distributional overlap among classes. The photometric bands (u, g, r, i, z) similarly display class-specific distributional patterns, though with varying degrees of separation. This heterogeneity in feature importance and distributional characteristics provides a favorable scenario for discriminant analysis methods, which explicitly model class-conditional distributions and can leverage these statistical differences to establish effective decision boundaries.

We employ three supervised learning methods to address this classification problem: LDA, QDA, and KNN. These methods are particularly well-suited for multi-class problems, as they naturally extend to handle more than two classes without requiring significant modifications. LDA assumes linear decision boundaries and identical covariance structures across classes, while QDA allows class-dependent covariance modeling, enabling more flexible decision boundaries. KNN provides a useful contrast to these parametric discriminant approaches, as it makes no distributional assumptions and adapts naturally to local feature patterns. While machine learning techniques such as neural networks and ensemble methods have been extensively applied to astronomical classification tasks [3]–[5], studies specifically exploring LDA and QDA for this problem remain relatively scarce.

The primary objective of this work is to evaluate how these methods perform given the heterogeneous feature importance and distributional characteristics identified in photometric data. By comparing linear and non-linear discriminant approaches alongside a non-parametric method, we aim to understand which modeling assumptions best leverage the statistical differences across classes. This analysis provides insights into the applicability of classical statistical methods for photometric classification of celestial objects.

II. METHODS

All methods evaluated in this study were implemented in Python using the scientific and machine learning ecosystem of the language. Data handling and preprocessing were performed with pandas, while model implementation and evaluation

relied on the scikit-learn library. Visualization of results, including confusion matrices and cross-validation curves, was carried out using matplotlib and seaborn.

A. Dataset

The dataset is a modified version of the SDSS DR17 classification dataset available on Kaggle [6]. During the preprocessing stage, all identification, positional, and instrumental metadata were removed, as these attributes do not contribute directly to the photometric classification task. Only the variables relevant for supervised learning were retained, namely the photometric measurements, redshift information, and the corresponding object class labels.

The final dataset has $N = 92303$ samples distributed across $L = 3$ classes: galaxies (56.1%), stars (23.4%), and quasars (20.5%). This class distribution reveals a moderate degree of imbalance, with galaxies constituting the majority class. Such imbalance can bias classifiers toward the dominant class, potentially inflating overall accuracy while degrading performance on minority classes.

The predictor variables consist of $D = 6$ features: five photometric magnitudes (u, g, r, i, z) corresponding to different wavelength bands in the SDSS filter system, and spectroscopic redshift. Table I provides detailed descriptions of each variable and their physical interpretation.

TABLE I
DESCRIPTION OF THE DATASET VARIABLES.

Tag	Unit	Description
u	—	Magnitude in the ultraviolet filter.
g	—	Magnitude in the green filter.
r	—	Magnitude in the red filter.
i	—	Magnitude in the near-infrared filter.
z	—	Magnitude in the infrared filter.
redshift	—	Measure of cosmological distance and recessional velocity.
class	—	Object label for classification.

B. Preprocessing and Data Spending

Prior to model training, some preprocessing steps were applied to the dataset. Outlier removal was performed using the empirical method proposed by Beck et al. [7], applied exclusively to samples classified as galaxies. This choice preserves the original photometric distributions of stars and quasars while mitigating extreme values within the dominant class.

To reduce skewness in the predictor variables, the Yeo–Johnson transformation was applied, followed by feature normalization. Both procedures were implemented using Scikit-Learn. After preprocessing, the data were organized into a predictor matrix $\mathbf{X}$, containing the photometric bands and redshift, and a label vector y , representing the object classes.

The dataset was subsequently split into training (70%) and testing (30%) sets using Scikit-Learn’s `train_test_split` function, with random shuffling and a fixed random seed to ensure reproducibility. All classification models were trained and evaluated using the same preprocessed data and partitioning.

C. Bayes Classifier and Discriminant Analysis

The ideal Bayesian classifier assigns an observation $\mathbf{x} \in \mathbb{R}^D$ to the class C_k that maximizes the posterior probability $P(y = C_k | \mathbf{x})$. By Bayes’ theorem, this posterior can be written as

$$P(y = C_k | \mathbf{x}) \propto p(\mathbf{x} | C_k) \pi_k$$

where $p(\mathbf{x} | C_k)$ is the class-conditional probability density function and $\pi_k = P(C_k)$ is the a priori probability of class C_k . In practice, π_k is typically estimated as the relative frequency of samples belonging to class C_k in the training data [8].

To make the Bayesian classifier easier, discriminant analysis methods assume that the class-conditional densities follow a multivariate Gaussian distribution,

$$p(\mathbf{x} | y = C_k) = \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k),$$

fully characterized by a mean vector $\boldsymbol{\mu}_k$ and a covariance matrix $\boldsymbol{\Sigma}_k$. These parameters are estimated from the training set using sample means and sample covariances computed separately for each class. The discriminant analysis methods evaluated in this work were implemented using the `LinearDiscriminantAnalysis` and `QuadraticDiscriminantAnalysis` classes from the `sklearn.discriminant_analysis` module.

D. LDA and QDA

Linear Discriminant Analysis is obtained by imposing the additional assumption that all classes share the same covariance matrix, i.e.,

$$\boldsymbol{\Sigma}_k = \boldsymbol{\Sigma}, \quad \forall k.$$

Under this assumption, taking the logarithm of the Bayesian decision rule and discarding terms independent of the class leads to a linear discriminant function of the form

$$\delta_k(\mathbf{x}) = \log \pi_k + \mathbf{x}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_k - 0.5 \boldsymbol{\mu}_k^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_k.$$

Classification is performed by assigning $\mathbf{x}$ to the class that maximizes $\delta_k(\mathbf{x})$. Since the discriminant function is linear in $\mathbf{x}$, the resulting decision boundaries between classes are hyperplanes in the feature space.

Quadratic Discriminant Analysis relaxes the assumption of a common covariance matrix and allows each class to have its own covariance,

$$\boldsymbol{\Sigma}_k \neq \boldsymbol{\Sigma}_j, \quad \text{for } k \neq j.$$

Applying the same logarithmic transformation to the Bayesian classifier yields the quadratic discriminant function

$$\delta_k(\mathbf{x}) = \log \pi_k - 0.5(\mathbf{x} - \boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k) - 0.5 \log |\boldsymbol{\Sigma}_k|.$$

As a consequence of the class-dependent covariance matrices, QDA produces nonlinear (quadratic) decision boundaries,

enabling greater flexibility in modeling class-specific dispersion patterns.

Both LDA and QDA are naturally formulated for multi-class classification problems, as the discriminant functions are defined independently for each class and compared simultaneously. LDA has a relatively low parameter complexity, requiring the estimation of one shared covariance matrix and one mean vector per class, which typically results in lower variance but higher bias when the equal-covariance assumption is violated. In contrast, QDA requires estimating a separate covariance matrix for each class, significantly increasing the number of parameters. This added flexibility reduces bias but increases variance and the risk of overfitting, particularly in high-dimensional settings or when class sample sizes are limited.

Given the heterogeneous distributional characteristics of the features and the dominant role of redshift in class separation [3], it is expected that models capable of capturing class-dependent covariance structures may achieve improved discrimination performance. In particular, QDA may benefit from its increased flexibility in modeling distinct dispersion patterns across classes, while LDA is expected to provide more stable performance due to its lower variance.

E. KNN

The K-Nearest Neighbors algorithm is a non-parametric, instance-based classification method that assigns a class label to a sample $\mathbf{x} \in \mathbb{R}^D$ based on the predominant class among its k nearest neighbors in the feature space. Proximity between samples is quantified through a distance metric $d(\mathbf{x}, \mathbf{x}_i)$, which defines the notion of neighborhood and directly influences the classifier's behavior.

In this work, similarity between samples is measured using the Euclidean distance, defined as

$$d(\mathbf{x}, \mathbf{x}_i) = \|\mathbf{x} - \mathbf{x}_i\|_2 = \sqrt{\sum_{j=1}^D (x_j - x_{i,j})^2},$$

where D denotes the dimensionality of the feature space. The Euclidean metric is well suited to continuous-valued, normalized features, as is the case for the photometric magnitudes and redshift used in this study.

KNN performs classification by identifying the set $N_k(\mathbf{x})$ of the k closest training samples according to the chosen distance metric and assigning $\mathbf{x}$ to the class that appears most frequently within this neighborhood. As a result, the classifier implicitly models complex, nonlinear decision boundaries that adapt to the local structure of the data, without relying on heavy assumptions about class-conditional distributions.

The KNN classifier was implemented using the `KNeighborsClassifier` from the Scikit-Learn library. To select an appropriate neighborhood size, a 5-fold cross-validation procedure was conducted for values of k ranging from 1 to 10. The results show that $k = 3$ achieves the highest average validation accuracy (0.9679), while larger values of k lead to a monotonic decrease in performance, as

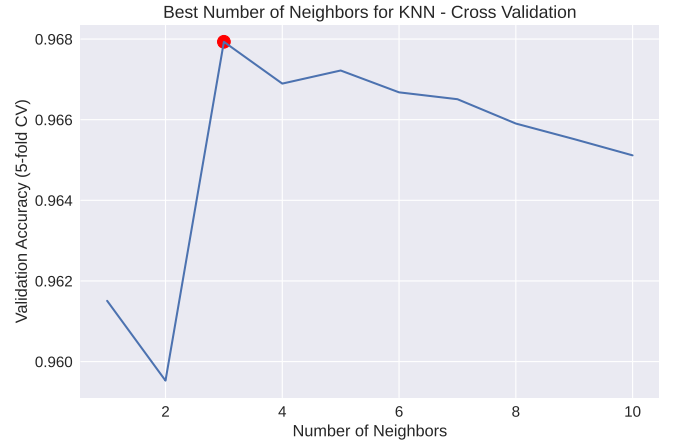


Fig. 1. Mean cross-validation accuracy as a function of the number of neighbors k (1–10) obtained using 5-fold cross-validation. The results indicate optimal performance at $k = 3$, followed by a monotonic decrease in accuracy for larger values of k .

illustrated in Fig. 1. This behavior reflects the bias–variance trade-off inherent to KNN where small values of k capture local structure excessively, whereas larger values smooth class boundaries, reducing discriminative power.

F. Evaluation

The performance of the classification models was assessed using the confusion matrix and overall classification accuracy. These evaluation tools provide complementary insights into both global performance and class-specific misclassification patterns.

For a multi-class classification problem with L classes, the confusion matrix is an $L \times L$ table in which each entry (i, j) represents the number of samples belonging to the true class C_i that were predicted as class C_j . Correct classifications correspond to the diagonal elements of the matrix, while off-diagonal entries represent misclassifications. In the context of this study, misclassification errors include, for example, galaxies incorrectly classified as stars or quasars, and quasars misidentified as galaxies or stars. Analyzing these errors allows for the identification of systematic confusion between specific object classes.

Overall classification performance was quantified using accuracy, defined as the proportion of correctly classified samples relative to the total number of samples,

$$Accuracy = \frac{\sum_{k=1}^L TP_k}{N}$$

where TP_k denotes the number of correctly classified samples of class k , L is the number of classes, and N is the total number of samples.

This metric provides a global summary of classification performance by equally weighting all samples, regardless of class membership. While simple and intuitive, accuracy may be misleading in the presence of class imbalance, as high accuracy can be achieved by favoring the majority class [?].

All evaluation metrics were computed using the `confusion_matrix` and `accuracy_score` functions from the `sklearn.metrics` module, ensuring consistency and reproducibility across experiments.

III. RESULTS

This section presents a comparative analysis of the classification results obtained by the evaluated methods. The discussion is based on the confusion matrices and on the overall classification accuracy.

A. Models' Performance

1) *LDA*: The confusion matrix obtained with LDA, shown in Fig.2, reveals substantial misclassification between galaxies and stars, as well as noticeable confusion between galaxies and quasars. These errors are consistent with the strong assumptions imposed by LDA, particularly the use of linear decision boundaries and a shared covariance matrix across classes. As a result, LDA achieved the lowest accuracy among the evaluated methods, with an overall value of 0.83.

2) *QDA*: The QDA results, illustrated in Fig.3, show a significantly improved diagonal structure in the confusion matrix, indicating a higher rate of correct classifications across all classes. By removing the equal-covariance assumption and allowing class-specific covariance matrices, QDA better captured the heterogeneous dispersion patterns present in the data. This increased flexibility led to a substantial performance gain, with an accuracy of approximately 0.95, while residual errors remained primarily associated with galaxy–quasar confusion.

3) *KNN*: The 3-Nearest Neighbors (3-NN) classifier produced the most concentrated diagonal structure in its confusion matrix, as shown in Fig.4, indicating the lowest overall misclassification rate. With an accuracy of 0.97, 3-NN outperformed both discriminant analysis methods. Its errors were sparse and largely limited to confusion between galaxies and quasars, reflecting intrinsic overlap in the feature space rather than model inadequacy.

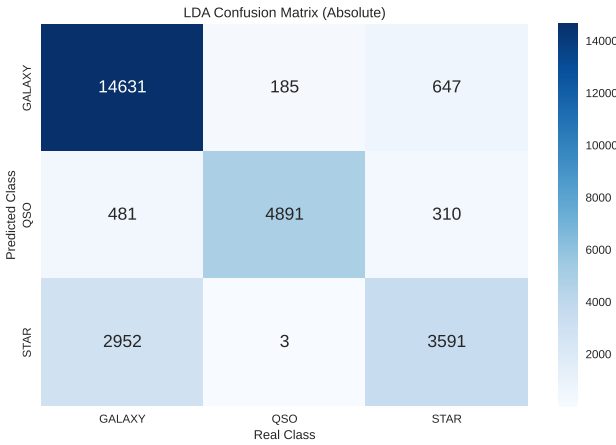


Fig. 2. Confusion matrix obtained with Linear Discriminant Analysis, showing significant misclassification between galaxies, stars, and quasars due to the limitations of linear decision boundaries.

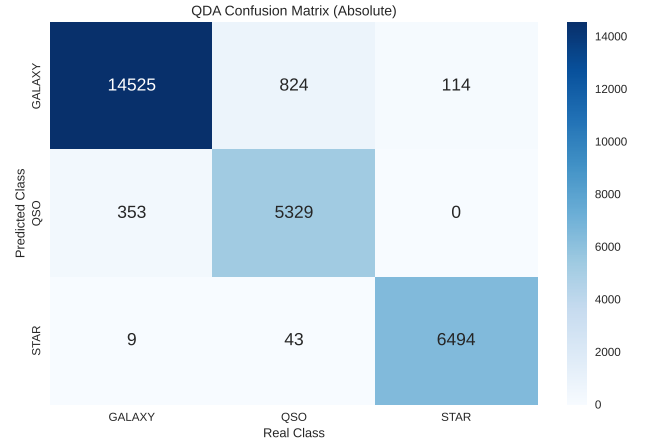


Fig. 3. Confusion matrix obtained with Quadratic Discriminant Analysis, illustrating improved class separation through class-dependent covariance modeling and reduced off-diagonal errors.

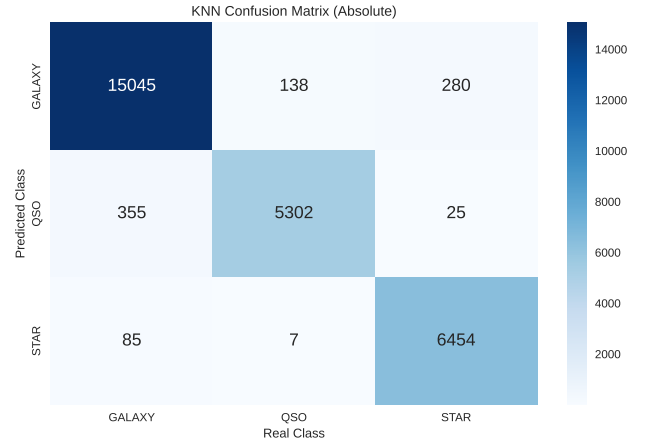


Fig. 4. Confusion matrix obtained with the 3-NN, highlighting strong diagonal dominance and minimal class confusion enabled by nonlinear, distance-based decision rules.

B. Conclusion

Comparing linear and nonlinear approaches, the results clearly demonstrate that nonlinear decision structures significantly improve classification performance for this dataset. LDA, constrained by linear boundaries, was unable to adequately separate classes with overlapping distributions, whereas both QDA and 3-NN benefited from increased model flexibility. Furthermore, the comparison between parametric models (LDA and QDA), which explicitly assume Gaussian class-conditional distributions, and the nonparametric KNN classifier, which makes no distributional assumptions, highlights an important trade-off. While QDA effectively exploits statistical structure when its assumptions are approximately satisfied, KNN achieves superior performance by adapting directly to the local geometry of the data, at the cost of higher sensitivity to noise and feature scaling [8].

A consolidated comparison of the classification accuracies

for all models is presented in Table II, which summarizes the performance differences observed across methods. Overall, this study demonstrates that accounting for nonlinear structure and class-dependent variability is essential for effective photometric classification of astronomical objects. The results confirm that both classical nonlinear statistical methods and nonparametric approaches remain highly competitive and relevant for multiclass problems in computational intelligence and astronomical data analysis.

TABLE II
CLASSIFICATION ACCURACY OBTAINED FOR EACH EVALUATED MODEL.

Model	Accuracy
LDA	0.83
QDA	0.95
3-NN	0.97

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